ECO3063: Computational Methods in Macroeconomics

Coursework 2

Due: 16:00, 8/5/2026

- A group may have **up to three** members (i.e. 1-3).
- Each group must submit **one solution**. Individual submissions are not required. If contributions are unequal, the group must include a brief statement describing each member's contribution as percentages that sum to 100%.
- The submission must include **four files**:
  1. One **code** file (or one zip-file containing the code; any programming language),
  2. One **document** file (containing all discussion, results, tables/figures, and references to the code/algorithm),
  3. One **spreadsheet** file (containing all data values used for Q2 and their sources, including: (i) an explanation of where each number was obtained, (ii) a screenshot of the raw data for each number, and (iii) the corresponding source URL), and
  4. One **video** file (see below).
- The code and the document must be self-contained, meaning:
  - The document includes everything needed to understand the approach and results. It must include a specific step-by-step numerical procedure (algorithm) that the code implements. If the document is missing a table or figure that is necessary, it will have a significant impact on the mark.
  - The code runs without requiring any missing files. If external libraries are used, the code must list them clearly and explain how to install them.
  - The code must contain detailed explanatory comments that correspond directly to the algorithm described in the document file. Comments are graded.
  - The code must be submitted as a separate file. Do not paste the code into the document file. The instructor must be able to run the code file.
- The **video** should be around 15 minutes, but it may be longer if necessary. While showing your code and document on screen, you must explain: (i) what each part does and what it reports, (ii) why you did it that way (for example, what you tried first and why you chose your final approach), with particular detail on any code techniques not covered in class, and (iii) what you were unable to achieve, including any failed attempts. Your explanation is particularly

important for **Q2, Q3, Q4**. The presentation may be delivered by the full group or by a single member representing the group. The title should start with '**Group "your group number"**'. See the instructions for how to create/upload your recording.

- Uploading a video file may take longer than uploading a standard submission file, and the upload time will depend on your internet connection and the file size. Please allow plenty of time for the upload.
- **Use of AI**: You may use AI while developing your solution, but the final submission must be original work produced by the group and must not be AI-generated. You are fully responsible for the content of the code and the document and must understand it completely. You must clearly demonstrate this understanding in the video. To ensure compliance, you may be required to explain your submission in person.
- Answer **all** questions.
- Add up the Student ID numbers of the group members. The last digit of the sum is your coursework number X.

1 Calibration and Welfare Analysis [10%]

(a) [5%] In Tutorial 4, we wrote code that replicates the calibration exercise for the Prescott economy. Use that code to update the "Labour Supply (Predicted)" and "Differences" columns of the table on slide 36 of Topic 4 lecture note. Discuss the results.

(b) [5%] Take the UK economy in 1993-1996 as a baseline. Suppose the UK's effective tax rate τ is changed from its baseline value of 0.44 to an alternative value taken from each of the other countries listed in the table. Compute the counterfactual hours worked and the implied welfare gains/losses as measured by the Consumption Equivalent Variation (CEV) to complete the following table. Discuss the results.

Counterfactual and Welfare Analyses for the UK Economy in 1993-96

Alternative Tax system	τ	Counterfactual Labour Supply (hours)	Welfare Changes (ε)
Germany	.59		
France	.59		
Italy	.64		
Canada	.52		
Japan	.37		
United States	.40		

2 Prescott in the 21st Century [30%]

In this exercise, you will make predictions for the working hours of the G7 countries for the year $201X$, where X is your coursework number.

(a) [20%] Add a new panel for the year $201X$ to the table on slide 36 of Topic 4 lecture note and fill in the columns for (i) "Labour Supply (Actual)", (ii) "Tax Rate τ", and (iii) "Consumption/Output (c/y)". Describe clearly and step-by-step how you compute each of these three objects and how each underlying number is obtained. Labour supply is measured as average weekly hours worked per person aged 15–64. You may estimate this as average weekly hours worked per employee multiplied by the labour force participation rate for ages 15-64. Prepare a spreadsheet in the required format (per the instructions above).

(b) [5%] Calibrate α by targeting hours worked in 1993-96 and $201X$ (and dropping 1970-74 from the calibration target). Report α and add "Labour Supply (Predicted)" and "Differences" to the table. Discuss whether the Prescott model is a good model for predicting labour supply in the 21st century, based on your results.

(c) [5%] In mapping the model to the data, we had to make several assumptions in order to measure the model objects. Identify one assumption that you consider particularly strong and explain why. Propose and briefly discuss one possible remedy.

3 Generalisation of the Prescott Model [25%]

The Prescott model matches hours worked in the G7 countries well for 1993–1996, but fits less well for 1970–1974. In this exercise, you will try to improve the fit by generalising preferences. Assume $V(100-h) = \alpha\frac{(100-h)^{1-\sigma}}{1-\sigma}$, where $\sigma > 0$ governs the curvature of leisure utility.

(a) [5%] Derive the first-order condition that characterises the optimal hours worked, which is a nonlinear equation of h involving $(\alpha, \sigma, \theta, \tau, c/y)$.

(b) [15%] Write a numerical algorithm to calibrate this economy by jointly choosing (α, σ) so that the model matches hours worked in 1993-1996 and 1970-1974. Report (α, σ) and explain how you verify that your reported (α, σ) is the correct solution. Report the results in a table as before and in figures as in slides 37-38 of Topic 4 lecture note. State whether the model fit improves relative to the baseline calibration, and briefly discuss why.

(c) [5%] Can the model be made to match the data exactly? If so, explain what you would need to change in the model and/or calibration procedure to achieve an exact fit, and discuss whether doing so would improve the Prescott model's usefulness for predicting hours worked across countries and over time. If you think an exact fit is not possible, explain why.

4 Ramsey Growth Model [25%]

Consider the Robinson Crusoe economy with total factor productivity (TFP). The production function is given by $F(z_t, k_t) = z_t k_t^\alpha$, with $\alpha > 0$. Consider a deterministic sequence of TFP such that

$$z_t = \sum_{j=0}^{t-1} \left(\frac{1}{2 + X/10} \right)^j, \text{ for } t = 1, 2, 3, \cdots,$$

where X is your coursework number. (Equivalently, $\{z_1, z_2, z_3, \cdots\} = \{1, 1+q, 1+q+q^2, \cdots\}$ with $q = \frac{1}{2+X/10}$).

(a) [5%] Derive the Euler equation for this economy. Solve for the steady-state values z^* and k^*.

(b) [5%] Write down an algorithm to solve the optimal capital path.

(c) [15%] Set $(\alpha, \beta, \delta, k_1) = (0.3, 0.96, 0.1, 1)$. Assume that the economy converges to the steady state by period $T = 30$. Write code for the function $K_T(k_2, k_1; \{z_t\}_{t=1}^T)$, defined analogously to $K_T(k_2, k_1)$ on slide 27 of Topic 5 (see also Tutorial 5). Your code must handle cases where $k_{t+2} < 0$ or $c_{t+1} < 0$. Plot the optimal capital sequence $\{k_t\}_{t=1}^T$ and discuss the results, including the shape of the capital path.

5 Visualisation [10%]

Find one figure from a professional article that may be misleading or unnecessarily cluttered. In your document file, include:

- The figure (as an image) and the source (e.g. URL, bibliographic information).
- An explanation of why the original figure may be problematic.
- An improved version of the figure that presents the same message more clearly.
- An explanation of why your version is better.

To obtain the underlying data, you may extract rough numbers by using AI or by eyeballing.